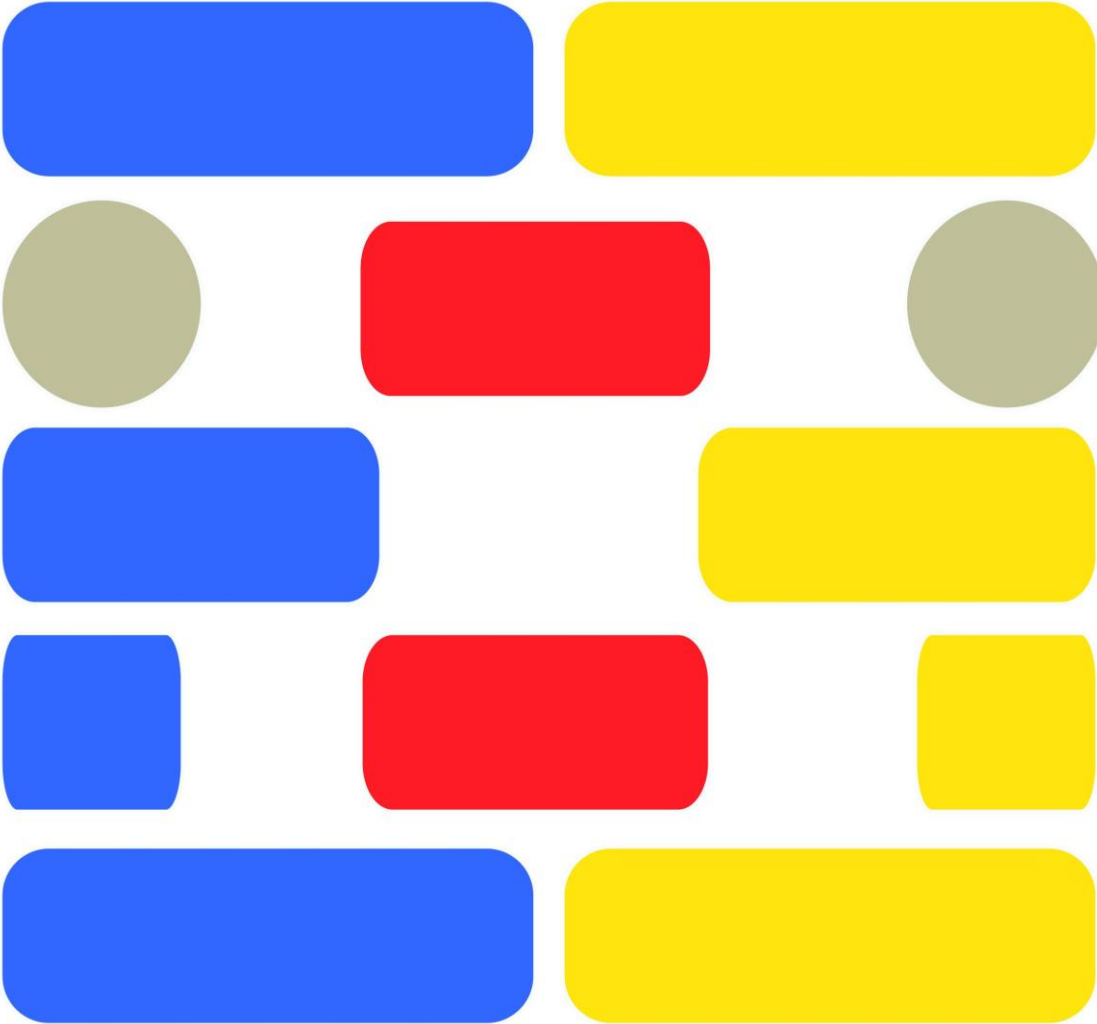




EC3 Built Environment – Ontology Lookup Service (BE-OLS)

Modelling and Standards Committee
16 June 2025



European Council on Computing in Construction (EC3)

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EC³

European Council on Computing in Construction

<https://ec-3.org/>

About EC³

The **European Council on Computing in Construction (EC³)** is a recently established society of construction professionals, academics, researchers and national Professional Bodies, aspiring to become the leading European forum in the area of information technology in construction engineering and management.

The European Council on Computing in Construction (EC3) advances professional knowledge and improves engineering practice in the built environment by fostering research, education and policy in current and emerging computing and information technologies. In order to achieve this societal goal, the association organises all





EC3 – M&S Committee

<https://ec-3.org/governance/technical-committees/modelling-standards-committee/>



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Modelling & Standards Committee

Modelling & Standards Committee

The Modelling & Standards (M&S) committee is a permanent committee of the EC³ and reports to the Board. This committee has the responsibility for promoting the research and innovation agenda of EC³ on all matters relevant to modelling and standards development.

A key output of the M&S committee has been the [BIM Standards Explorer](#), a Dashboard of Standards that are relevant to the EU construction industry.

Members of Project D



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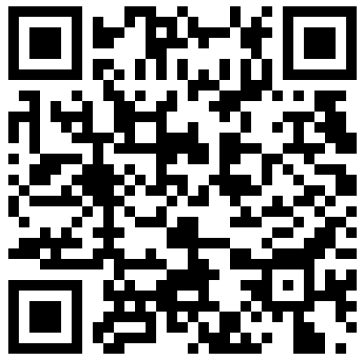


EC3 – BE-OLS:

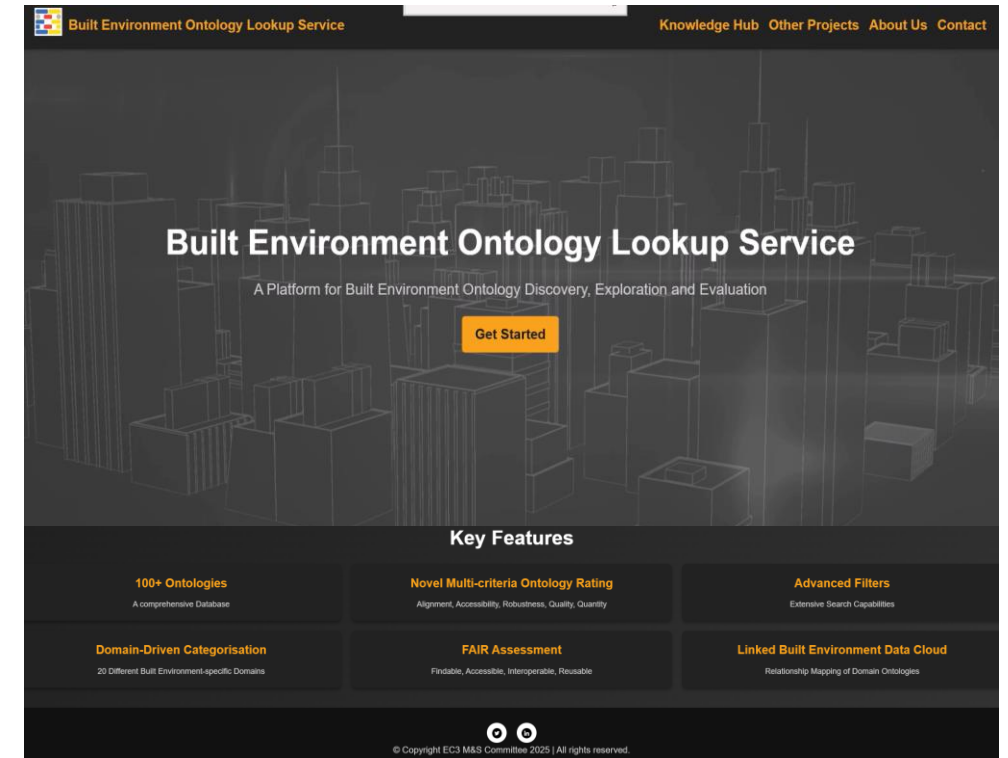
Answering the ontology (re)use gap

Some ontologies are:

- Developed as **outputs of short-term projects**
- Developed as tools toward better **data integration in a bigger project**
- Not always designed for reuse by others **outside the original project**
- Not easily **discoverable** (to be reused)



<https://cyberbuildlab.github.io/BE-OLS/index.html>





EC3 – BE-OLS

- > 100 Ontologies

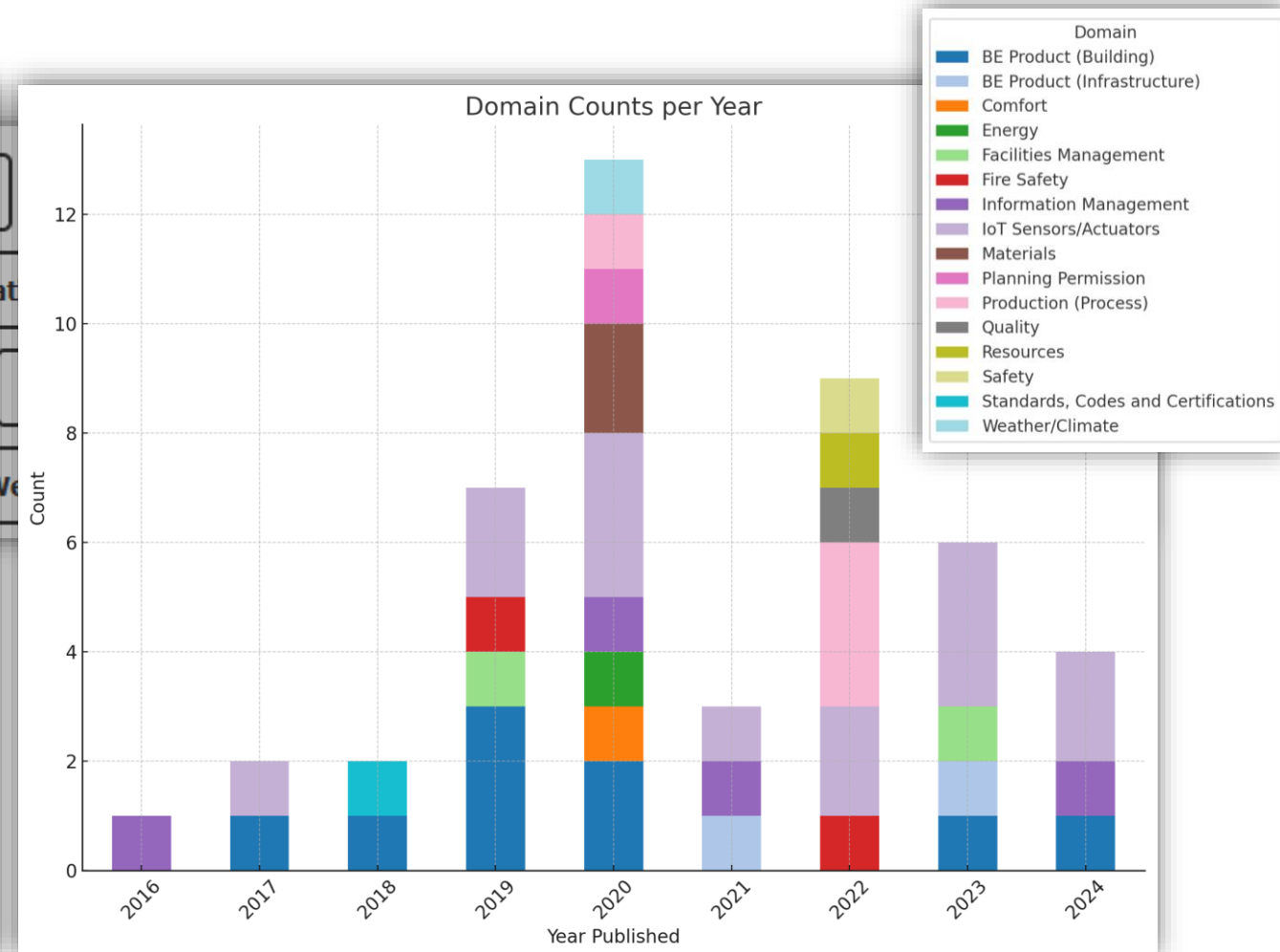
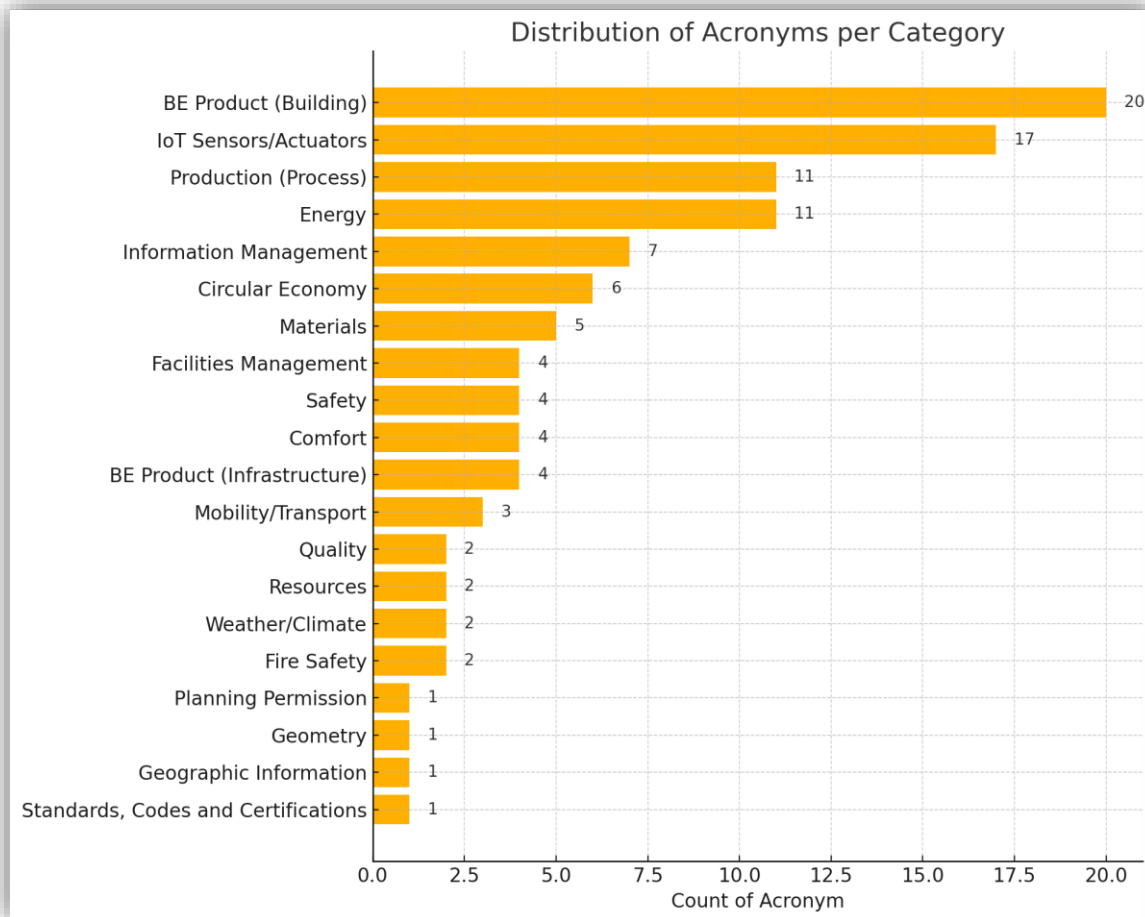
Ontology Name	Acronym	Primary Domain	Secondary Domain	FAIR Score	Year
Steel Element Reuse Ontology	SERO	Primary Domain: Circular Economy	Secondary Domain: Materials	FAIR Score: N/A	2024
Semantic Sensor Network Ontology - SOSA	SOSA	Primary Domain: IoT Sensors/Actuators	Secondary Domain: N/A	FAIR Score: N/A	2017
Semantic Sensor Network Ontology - SSN	SSN	Primary Domain: IoT Sensors/Actuators	Secondary Domain: N/A	FAIR Score: N/A	2017
Scan To Graph	STG	Primary Domain: BE Product (Building)	Secondary Domain: Remote Sensing	FAIR Score: N/A	2019
Users and Preference Information	THACTOR	Primary Domain: Comfort	Secondary Domain: N/A	FAIR Score: N/A	2013
Architecture and Building Physics Information Ontology	THBUILDAP	Primary Domain: BE Product (Building)	Secondary Domain: N/A	FAIR Score: N/A	2013
Energy and Resource Information	THENERIS	Primary Domain: Energy	Secondary Domain: N/A	FAIR Score: N/A	2013
User Behavior and Building Process Information	THPROCESS	Primary Domain: Energy	Secondary Domain: N/A	FAIR Score: N/A	2013
Weather and Exterior Influence Information	THWEATHER	Primary Domain: Weather/Climate	Secondary Domain: N/A	FAIR Score: N/A	2013
User Navigation Ontology	UNO	Primary Domain: Mobility/Transport	Secondary Domain: N/A	FAIR Score: N/A	2009
User Profile Ontology	UPO	Primary Domain: Mobility/Transport	Secondary Domain: N/A	FAIR Score: N/A	2009
BIMERR Weather Ontology	WEAT	Primary Domain: Weather/Climate	Secondary Domain: N/A	FAIR Score: N/A	2020
Facility Management Building Energy Ontology for Renovative Design Decisions	FM-ENER	Primary Domain: Facilities Management	Secondary Domain: Energy	FAIR Score: N/A	2022
GIS-based ontology for surrounding buildings to support building renovation	N/A	Primary Domain: BE Product (Building)	Secondary Domain: N/A	FAIR Score: N/A	2020



EC3 – BE-OLS:

Domain and Sub-domain Analysis

- > 100 Ontologies

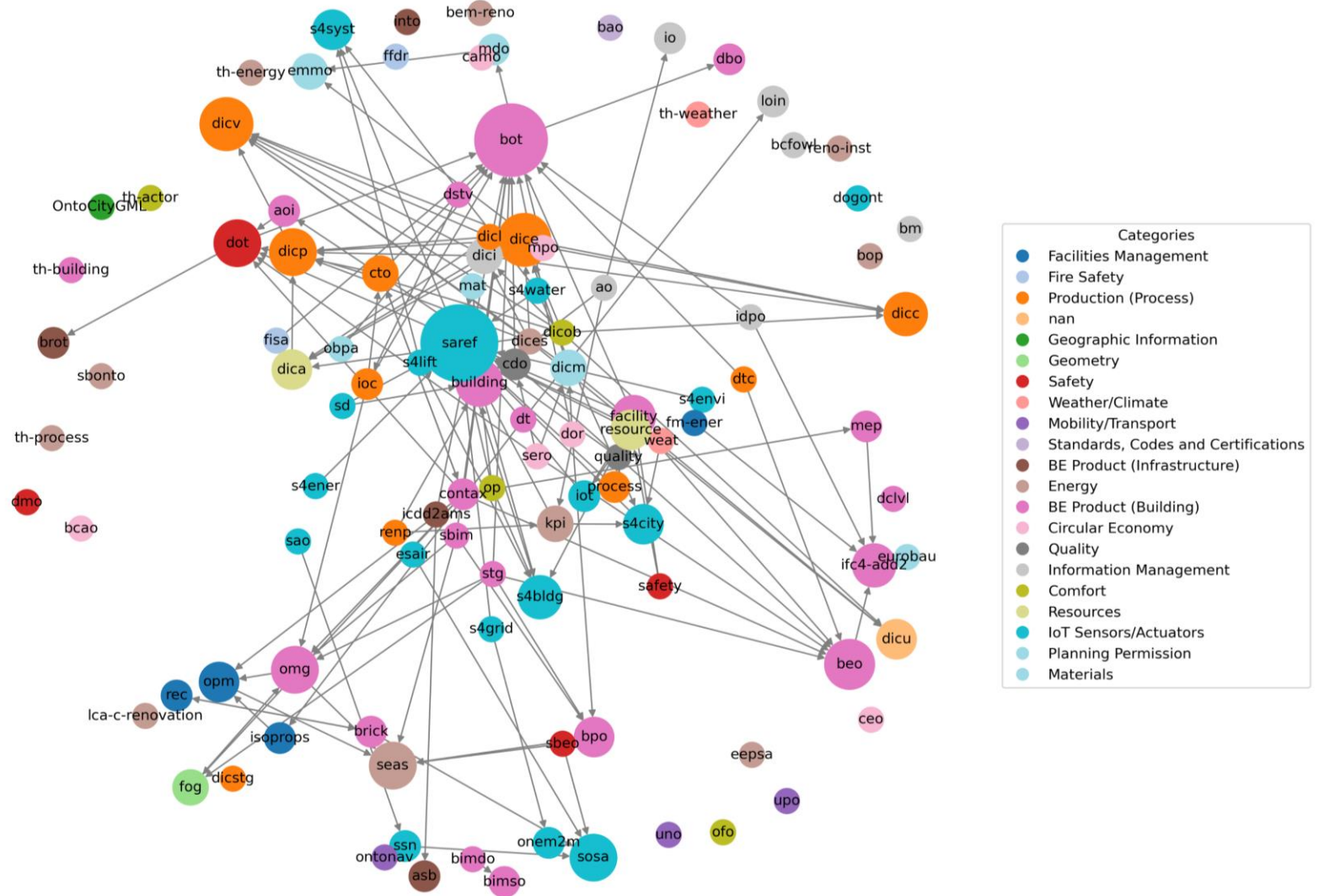




EC3 – BE-OLS:

Network Analysis of domain ontologies

- > 100 Ontologies
- 20 Sub-domains
- Ontology network





EC3 – BE-OLS:

Metadata and Enriched Metadata

- > 100 Ontologies
- 20 Sub-domains
- Ontology network
- Metadata

M&S Analysis

Metadata
From
Ontology's
URL or TTL

Enriched Metadata

Enriched Metadata

Enriched Metadata

BIMERR Annotation Objects Ontology (AO)

Field	Value
Name	BIMERR Annotation Objects Ontology
Acronym	AO
Year published	2020
Version	0.0.2
Licensing	CC BY 4.0
Short Description	The Annotation Objects ontology aims to represent the annotations produced during the development of a building renovation project. These annotations serve to inform about issues or missing information that could be relevant for the project, such as indicating that a building element is missing in the BIM model.
Primary Domain	Information Management
Reference	https://bimerr.iot.linkeddata.es/
Linked-to ontologies AECO	BUILDING, KPI, IO

Cluster Name
BIMERR



EC3 – BE-OLS:

Overview of existing frameworks for ontologies and Linked Data

2006

2006: 5-Star Linked Open Data
Proposed by Tim Berners-Lee to improve data availability and interoperability.

2012

2012: 5-Star Linked Data Vocabulary
Bernard Vatant focused on vocabulary quality and documentation.

2014

2014: 5-Star Vocabulary Use
Janowicz et al. proposed a new scale to evaluate vocabulary usage and referencing.

2016

2016: FAIR Principles
Wilkinson et al. introduced Findable, Accessible, Interoperable, Reusable (FAIR) data management guidelines.

2020

2020: FAIRsFAIR Recommendations
Hugo et al. published 17 recommendations and 14 practices for semantic artefacts.

2023-24

2023–2024: Onto4Reuse & O4OA Frameworks
Conceptual reuse tools based on foundational ontologies for semantic alignment.

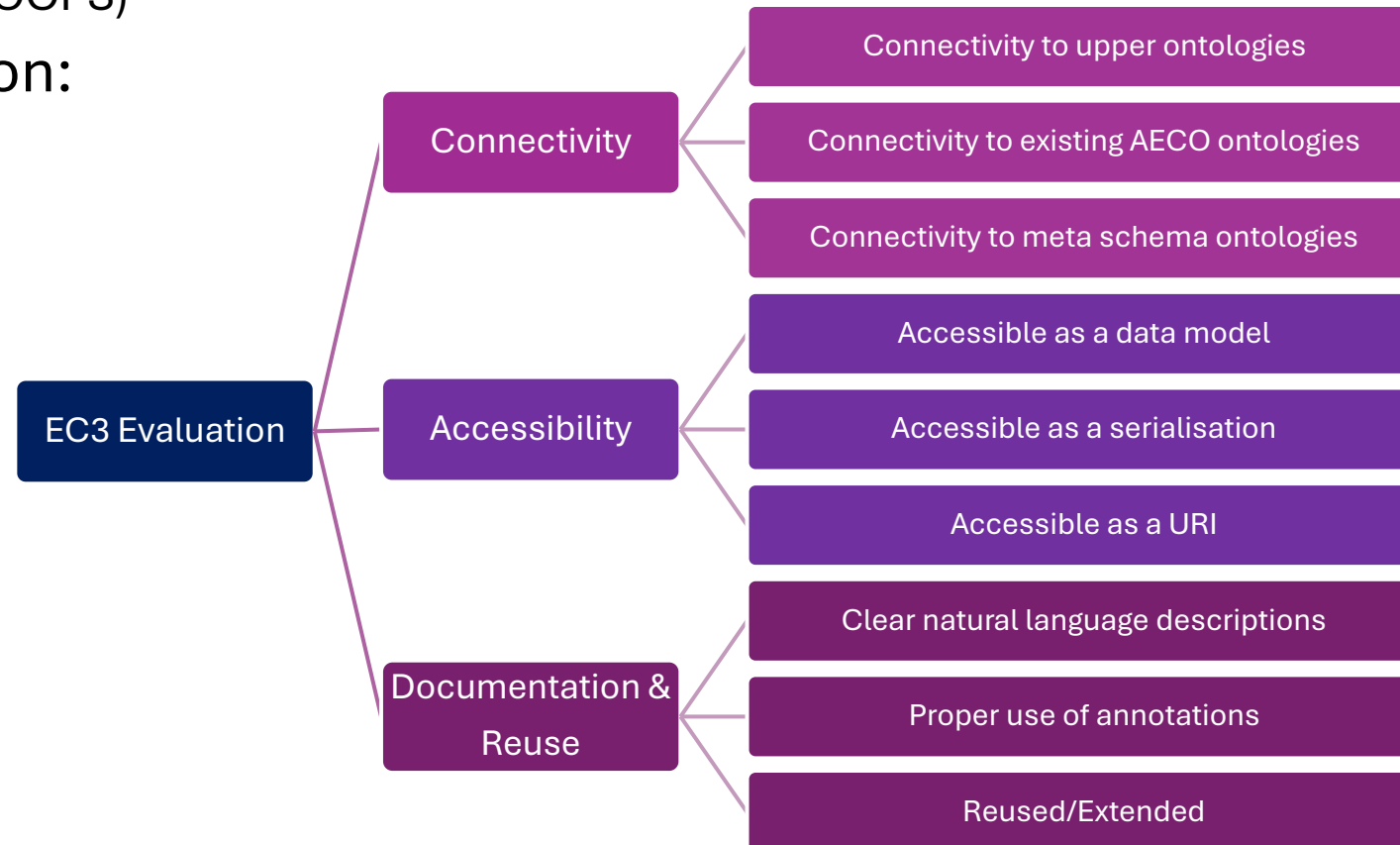


EC3 – BE-OLS:

A complementary evaluation framework for ontology (re)use

- Ontology evaluation:

- FAIR Score (FOOPS)
- EC3 evaluation:





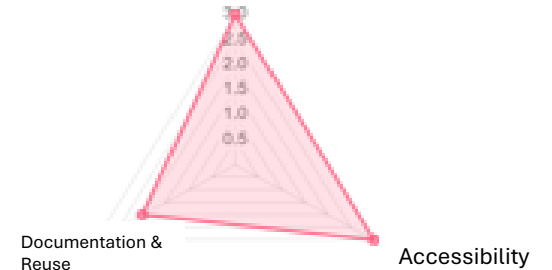
EC3 – BE-OLS:

A complementary evaluation framework for ontology (re)use

- Ontology evaluation:
 - FAIR Score (FOOPS)
 - EC3 evaluation:

Decommissioning and Reuse Ontology (DOR)	
Field	Value
Name	Decommissioning and Reuse Ontology
Acronym	DOR
Year published	2023
Licensing	CC BY 4.0
Used Standards	ISO/DIS 59004
Primary Domain	Circular Economy
Secondary Domain	Materials
Reference	https://orbi.lu.uni.lu/handle/10993/58846
Linked-to ontologies AECO	BOT, BPO, DICM

Decommissioning and Reuse Ontology
Connectivity



Evaluation Axis	Criteria	Presence	Score
Accessibility	Linkage to upper ontologies	No	3
	Linkage to existing AECO ontologies	Yes	
	Linkage to meta schema ontologies	Yes	
Connectivity	Conceptual Data model available	Yes	3
	Accessible as Serialization	Yes	
	Accessible as a URI	Yes	
Documentation & Reuse	Clearly documented	Yes	2
	Use of annotations	Yes	
	Reused/Extended	No	



EC3 – BE-OLS

-  Project Kick-off: October 2024
-  Team: 5 volunteers (academic)
-  Future work:
 - Completing missing ontologies
 - Enhancing information quality
 - *Completing evaluation axes: Quantity (ratio/density or number of classes, properties and relevant data) and Robustness (how it was evaluated, validated and ...?)*
 - Extracting metadata from ttl files (check the overlap of concepts)
 - Using LLMs





EC3 – BE-OLS:

Thanks for your attention!



Help us improve!
(Feedback Form)



Submit issues



GitHub



<https://github.com/CyberbuildLab/BE-OLS>



Join our team
Or, disseminate



<https://cyberbuildlab.github.io/BE-OLS/index.html>